Earthling Field Guide – Middle School Edition: Classroom Guide for Teachers

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A free browser game for grades 6–8 in which students play a friendly alien studying life on Earth. Across 26 levels in six worlds they learn the vocabulary and big ideas of the Pennsylvania middle school life science standards, in English or Spanish. This guide tells you what to expect in each level and how to help a student who is stuck. The game is a work in progress: it covers the Life Science standards now, and other science areas will come in future updates.

Getting started

> **Before class: ask your IT department to approve the game.** Many schools block websites on student Chromebooks until IT allows them (sometimes called whitelisting). Ask IT to allow these addresses, then test the game on a student Chromebook before the lesson:
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> - **jillelynx.github.io** (the game itself)
> - **cdnjs.cloudflare.com** (the 3-D graphics library the game loads)
> - **fonts.googleapis.com** and **fonts.gstatic.com** (the game's fonts; the game still works without them)

- **Open the game:** go to jillelynx.github.io/EarthlingFieldGuide_MS in any up-to-date web browser. Nothing to install and no accounts. It runs on school Chromebooks, laptops and desktops; tablets get on-screen buttons.
- **Saving:** progress saves automatically in that browser on that device. A student who moves to a different computer starts over there, so it helps if students use the same device each time. On shared computers, Settings has a button to erase progress on this device.
- **Language:** students choose English or Español on the main menu. Everything in the game switches, including questions and feedback.
- **The main menu:** 🎨 Customize your alien lets students design their explorer (body, color, eyes, mouth, antennae, gear). Then they press 🎮 Platformer Adventure to start, or Map to pick any level. Every world is open from the start, so you can assign the world that matches your unit.

- **Settings:** text size, low graphics (use this if a computer runs slowly), music, sound and sound captions.
- **Time:** most levels take 10–20 minutes; a world takes about a class period or two.
- **No typing:** there are no written answers. Students choose answers, sort cards, read charts and build models.

Controls

Action	Keyboard
Move	Left and right arrows, or A and D
Jump (press again in the air for a double jump)	Space, up arrow or W
Swim up in water	Hold jump
Duck, crawl under low spots, drill down	Down arrow or S
Run	Hold Shift
Scan, talk, use a station, answer	E or Enter
Choose an answer	A, B, C or D (or click it)
Pause	P or Esc

Tablets show on-screen buttons. The pause menu has Restart level, the Field Journal, Settings and Main menu.

How a level works

Every level is a real place (a Pocono vernal pool, Hawk Mountain, the Great Barrier Reef) or a trip inside a living thing with the alien's shrink ray. Students run and jump from left to right. Along the way:

1. **Scan things** (press E near a glowing object) to read a short fact card. New words go into the Field Journal.
2. **Stations** block the path until the student answers 2–4 multiple-choice questions. A wrong answer shows a short explanation and adds a few seconds to the clock. After two wrong tries the right answer glows, so nobody gets stuck for good.

3. **Activities** test the level's big idea: sorting cards into groups, putting steps in order, reading a chart ("Read the Data"), building a cell or model from parts, collecting only the right samples in a hunt (wrong samples cost points), keeping a body or ecosystem in balance, and finding the right path through a maze.
4. **Refuel stations** offer an optional bonus quiz for extra points. Students can leave any time (Esc).
5. **The Lab Lock** at the end asks questions using clues found in the level. Answering them opens the lock and gives a short debrief.
6. **The finish beam** ends the level and shows the report: stars, time, points and words found.

Some levels have a portal to a mini-game, a rare specimen to find, a boss, or extra hazards such as crumbling platforms, moving obstacles and water.

When a student is stuck

What the student sees	What to tell them
A glowing wall or gate blocks the path	It's a station. Stand on it, press E, and answer the questions. After two wrong tries the right answer glows.
Doesn't know an answer	Scan the glowing objects nearby (E) and open the Field Journal (pause menu). The fact cards hold the words and ideas the questions ask about.
Fell into the pit under a crumbling bridge	Walk to the right end of the pit, jump onto the block there, then jump up to the ground. Crumbling planks break if you stand still, so keep moving.
In the water and can't get out at the shore	Hold jump (do not tap) to swim all the way to the surface, then keep holding right to climb onto land. It takes a few seconds.
A low ceiling or narrow gap	Hold down to duck and crawl through.
A faint, see-through platform	It's a hidden platform that Lab goggles make solid (goggles come with 3 stars or from the station shop). The main path never needs them.

What the student sees	What to tell them
A maze with labeled doorways	Read the sign at the start of the maze. Only the doorway named for the next step opens; the others are dead ends that cost 25 points and push you back to try again.
A sample hunt	Collect only the samples that fit the rule at the top of the screen. Decoys cost points, so read each label first.
Build a cell or model	Pick up the parts that belong (E) and carry them to the model at the end. Wrong parts buzz and don't fit.
Put steps in order	Stand at the step that comes next and press E.
A balance game (a body or ecosystem gauge)	Use the two levers to push the reading back into the normal range shown on the gauge, and hold it there until the game says it is balanced.
Fossil dig in 6-1	Stand on soft rock and hold down to drill. If the drill gets too hot, wait a moment for it to cool.
A portal mini-game	Follow the goal at the top of the screen. Esc leaves the mini-game.
Keeps losing lives	The student restarts at the last checkpoint flag. Encourage slower, shorter jumps. If the game stutters, turn on Low graphics in Settings.
Truly stuck or lost	Pause (P or Esc) and choose Restart level. Stars already earned are kept.

Tip: let students read the toasts (the short messages at the bottom of the screen). They explain every new obstacle the first time it appears.

World by world

Each world ends with a celebration screen and a printable certificate once every science level in it has at least one star.

World 1: What Living Things Are Made Of

Level	What students learn and do	Watch for
1-1 Pocono Vernal Pool	Living vs. nonliving things; all living things are made of cells. Specimen hunt (collect only things made of cells), microscope fact cards, a chart.	Decoys in the hunt cost points. A crawl space and a crumbling bridge.
1-2 Deep-Sea Vent	Sinking through the ocean zones to hydrothermal vents; one-celled vs. many-celled life.	Swimming: hold jump to rise.
1-3 Backyard Garden	Shrinking into a plant cell; what each cell part does for the whole cell. Cell-part maze; build a spinach leaf cell.	The maze uses the cell's parts; build only parts a plant cell has.
1-4 Rocky Tide Pool	Tide-pool animals as teams of cells; plant cells vs. animal cells.	Lab Lock at the end uses clues from the level.

World 2: The Body as a System

Level	What students learn and do	Watch for
2-1 Inside a Hiker's Body	Cells form tissues, tissues form organs, organs form systems. Maze; a portal to the Immune Defense mini-game.	Portal clues; Esc leaves the mini-game.
2-2 Hawk Mountain Trail	Body systems working together on a hike (circulatory, respiratory, muscular, nervous, digestive, excretory). Heart-rate chart; a body balance game.	Balance game: keep the gauge in the normal range.
2-3 Cherry Springs at Night	Senses: stimulus, receptor, nerve, brain. One of the darkest night skies in the East.	Dim scenes; turn up screen brightness if needed.

Level	What students learn and do	Watch for
2-4 Your Brain	Inside a giant neuron net (Franklin Institute): when the brain reacts right away and when it stores a memory. Reaction-time chart.	Climbing and crawl spaces.

World 3: Food and Energy for Life

Level	What students learn and do	Watch for
3-1 Jennings Prairie	Photosynthesis: plants use light, water and carbon dioxide to make sugar and give off oxygen. Oxygen chart; steps in order.	Read chart titles and units carefully.
3-2 Monterey Kelp Forest	Energy from the Sun enters a kelp forest food web; matter moves through it.	Swimming; a maze.
3-3 A Lunch's Journey	A sandwich's trip through the body: food molecules are broken apart and rebuilt into new molecules, releasing energy. Digestion Mini Golf portal.	Mini golf: match each food to the right part of the digestive system.
3-4 Garden Test Plots	Genes and the environment (light, water, space, food) both affect growth. Plant-mass chart; balance game.	Questions ask whether a difference comes from genes or the environment.

World 4: Ecosystems

Level	What students learn and do	Watch for
4-1 Great Barrier Reef	Predation, competition and mutualism, and the same patterns in other ecosystems.	Swimming. At the tall shore wall, hold jump to reach the surface.

Level	What students learn and do	Watch for
4-2 Allegheny National Forest	Acorn boom-and-bust years change mouse and deer populations. Population charts; a deer balance game.	Charts show changes a year apart.
4-3 Under a Fallen Log	Decomposers; matter cycles and energy flows; the boundary of a system.	Many crawl spaces (hold down); a maze.
4-4 Yellowstone After the Fire	How a change to an ecosystem affects populations, and which evidence supports a claim.	Evidence questions: pick data, not opinions.
4-5 Susquehanna River Valley	Comparing designs that protect biodiversity and ecosystem services (dams, riparian buffers, fish lifts). Boss: The Silt Storm.	The boss is a set of questions; answer them to clear the river.

World 5: Passing On Traits

Level	What students learn and do	Watch for
5-1 Bowman's Hill Wildflowers	How flower structures and animal behaviors help plants and animals reproduce. Bee-net experiment chart.	Behavior vs. structure sorting.
5-2 Strawberry Coast	One parent (asexual: identical offspring) vs. two parents (sexual: variation); Punnett squares.	Questions about copies vs. new combinations.
5-3 Cave of the Pale Fish	Mutations can be harmful, helpful or neutral. A dark limestone cave.	A dark cave with falling rocks, a maze and crawl spaces.
5-4 Farm and Research Lab	People shaping traits: selective breeding, genetic modification, gene therapy, and their benefits and concerns.	A benefits-or-concerns sort.

World 6: Change Over Time

Level	What students learn and do	Watch for
6-1 Canyon Fossil Dig	Rock layers as a record: deeper is older; patterns in the fossil record.	Drilling: hold down on soft rock; let the drill cool.
6-2 Whale Valley	Whale skeletons with legs: anatomy shows relationships between living and fossil animals.	A maze; a leg-bone chart.
6-3 Woods Hole Marine Lab	Comparing drawings of embryos to find hidden similarities.	Drawings are simplified, not to scale.
6-4 Daphne Major Finches	During the 1977 drought, beak size changed who survived; simple probability.	Questions about percentages and "how many times more likely."
6-5 The Selection Simulator	Natural selection in numbers: a peppered-moth simulator portal and graphs over generations. Boss: The Graph Gremlin.	The boss asks graph questions.

Bonus speed runs

The last space in each world on the Map is a square 👾 bonus level: an optional, just-for-fun 8-bit speed run with no science questions. Students collect coins, dodge obstacles and race the clock; stars come from the time. Bonus runs never count toward certificates, so they work well as a reward after a world is done. Each student plays as a pixel version of the alien they designed.

A glowing orb in each run teaches a new move, with a practice section right after it. Moves are kept for later bonus runs.

Bonus level	Habitat	New move	How to do it
1-5 Pond Edge	Pond	Ground Pound	In the air, press down: breaks bricks, squashes enemies
2-5 Desert Dash	Desert	Air Dash	In the air, press Shift (once per jump)

Bonus level	Habitat	New move	How to do it
3-5 Rainforest Run	Rainforest	Wall Jump	Jump toward a wall, then jump again while touching it
4-6 Tundra Sprint	Tundra (with sled runs)	Glide	While falling, hold jump
5-5 Reef Rush	Coral reef	Spin Attack	Press E
6-6 Penn's Woods	Pennsylvania forest in fall	All five	A power orb before each section

Each run ends with a boss, a giant virus. Stomp on it (or Ground Pound or Spin into it) 3 times to open the exit gate; touching its side costs a life.

Stars, progress and certificates

- **Stars:** 3 stars for 90% or more right on the first try, 2 stars for 70% or more, otherwise 1. Students can replay a level to raise their stars.
- **Progress screen:** stars, words discovered and rare specimens for each world.
- **Certificates:** finishing every science level in a world (at least 1 star each) opens a celebration screen with a printable certificate; finishing all six worlds earns a whole-game certificate. Students type their name only to print it; names are never saved.
- **Teacher view:** add `#teacher` to the end of the game's address (jillelynx.github.io/EarthlingFieldGuide_MS/#teacher) to see the standards reference sheet in About: every standard's exact wording, where it is taught and how many items cover it, with a Print button. Students never see standard codes.

What the game covers (a work in progress)

This version teaches the 21 Pennsylvania grades 6–8 **Life Science** standards (STEELS 3.1.6-8.A–U). The Grade 8 Science PSSA also tests Environmental Literacy and Sustainability, Physical Science, Earth and Space Science, and Technology and Engineering; those areas will be added in future updates. All questions were written for this game and are not official PSSA items.

Found an error, or have feedback or an idea? Please let the creator know on the game's feedback page: github.com/jillelynx/EarthlingFieldGuide_MS/issues (a free GitHub account is needed to post).